

Simultaneous language learning

```
knitr::opts_chunk$set(echo = TRUE)

#English analysis

file_path <- "~/Desktop/Simultaneous_learning.csv"
data <- read.csv(file_path)

data$Time <- factor(data$Time)
data$Type <- factor(data$Target.Control)
data$Categorical_score <- factor(data$Categorical_score, ordered = TRUE, levels = 1:2)
data$Language <- factor(data$Language)

data <- subset(data, Language == "English")

data$Time <- factor(data$Time)
data$Type <- factor(data$Target.Control)
data$Categorical_score <- factor(data$Categorical_score, ordered = TRUE, levels = 1:2)
data$Language <- factor(data$Language)

model_int <- glmer(
  Categorical_score ~ Time * Target.Control + English_proficiency + Dutch_proficiency +
    (1 | Participant) + (1 | ItemID),
  family = binomial, data = data
)
summary(model_int)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: Categorical_score ~ Time * Target.Control + English_proficiency +
  Dutch_proficiency + (1 | Participant) + (1 | ItemID)
Data: data
```

AIC	BIC	logLik	deviance	df.resid
616.0	655.5	-300.0	600.0	1032

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.9426	-0.3291	-0.1528	-0.0578	6.8360

Random effects:

Groups	Name	Variance	Std.Dev.
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```

ItemID      (Intercept) 1.262    1.123
Participant (Intercept) 1.238    1.113
Number of obs: 1040, groups: ItemID, 40; Participant, 13

```

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-17.05843755	3.87473368	-4.402	0.0000107 ***
Time2	-0.00002675	0.38569497	0.000	0.99994
Target.Control	0.13108811	0.50961555	0.257	0.79700
English_proficiency	0.15377859	0.05763789	2.668	0.00763 **
Dutch_proficiency	0.06086970	0.04016495	1.515	0.12965
Time2:Target.Control	0.14824143	0.47220569	0.314	0.75357

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	Time2	Trgt.C	Engls_	Dtch_p
Time2		-0.050			
Target.Cntrl	-0.088	0.378			
Englsh_prfc	-0.827	0.000	0.002		
Dtch_prfcnc	-0.200	0.000	0.001	-0.369	
Tm2:Trgt.Cn	0.036	-0.817	-0.468	0.003	0.002

```

r2_model_int <- performance::r2(model_int)
print(r2_model_int)

```

R2 for Mixed Models

Conditional R2: 0.637
Marginal R2: 0.361

```

emm <- emmeans(model_int, ~ Time * Target.Control, type = "response")
summary(emm)

```

Time	Target.Control	prob	SE	df	asympt.LCL	asympt.UCL
1	0	0.0309	0.0171	Inf	0.0103	0.0890
2	0	0.0309	0.0171	Inf	0.0103	0.0890
1	1	0.0350	0.0169	Inf	0.0135	0.0880
2	1	0.0404	0.0191	Inf	0.0157	0.0997

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

```

pairs(emm, by = "Target.Control") # change over Time within each group

```

Target.Control = 0:

contrast	odds.ratio	SE	df	null	z.ratio	p.value
Time1 / Time2	1.000	0.386	Inf	1	0.000	0.9999

Target.Control = 1:

contrast	odds.ratio	SE	df	null	z.ratio	p.value
Time1 / Time2	0.862	0.235	Inf	1	-0.544	0.5864

Tests are performed on the log odds ratio scale

```
pairs(emm, by = "Time")
```

```
Time = 1:
  contrast                                odds.ratio    SE  df null z.ratio p.value
Target.Control0 / Target.Control1         0.877 0.447 Inf    1  -0.257  0.7970
```

```
Time = 2:
  contrast                                odds.ratio    SE  df null z.ratio p.value
Target.Control0 / Target.Control1         0.756 0.384 Inf    1  -0.551  0.5819
```

Tests are performed on the log odds ratio scale

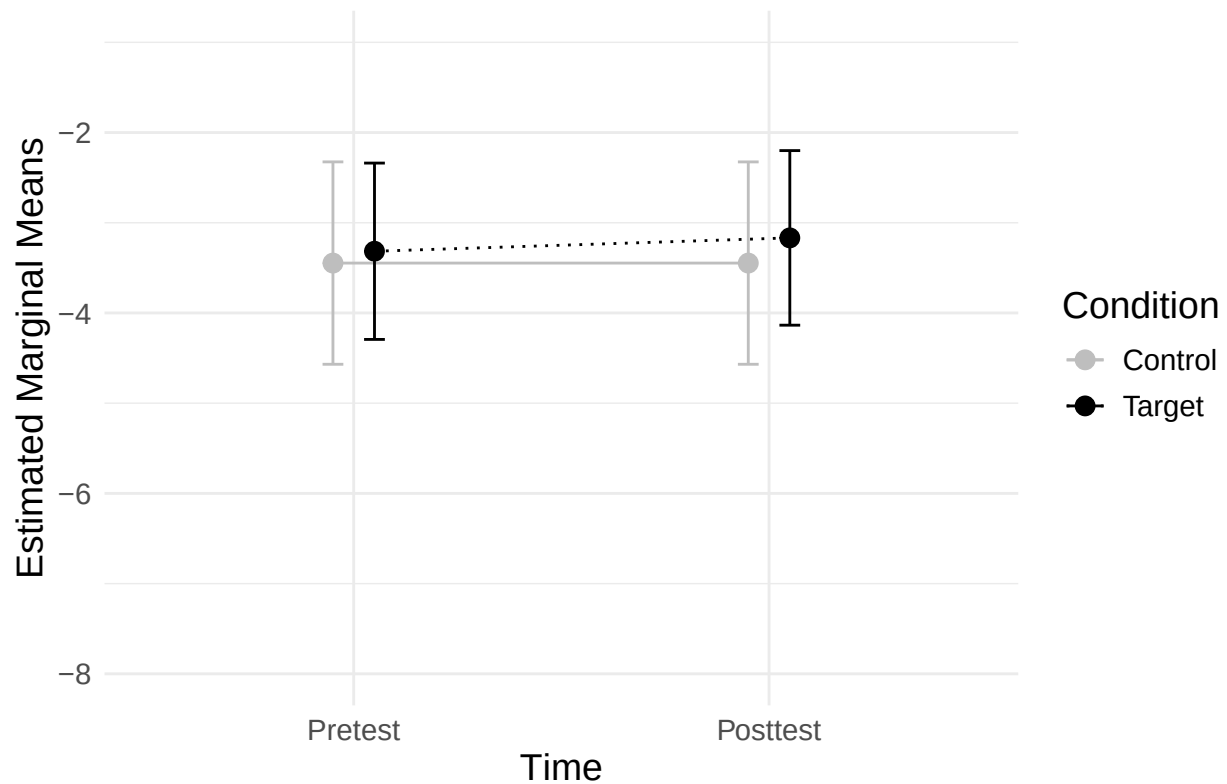
```
library(ggplot2)
```

```
emm <- emmeans(model_int, ~ Time * Target.Control)
marginal_means_df <- as.data.frame(emm)
```

```
marginal_means_df$Target.Control <- factor(marginal_means_df$Target.Control, labels = c("Control", "Target"))
marginal_means_df$Time <- factor(marginal_means_df$Time,
                                levels = c(1, 2),
                                labels = c("Pretest", "Posttest"))
```

```
ggplot(marginal_means_df, aes(x = Time, y = emmean, color = Target.Control, group = Target.Control)) +
  geom_point(size = 3, position = position_dodge(width = 0.2)) +
  geom_line(aes(linetype = Target.Control), position = position_dodge(width = 0.2)) +
  geom_errorbar(aes(ymin = asymp.LCL, ymax = asymp.UCL),
               width = 0.1, position = position_dodge(width = 0.2)) +
  labs(
    title = "(A) English learning outcomes",
    x = "Time",
    y = "Estimated Marginal Means",
    color = "Condition",
    linetype = "Condition"
  ) +
  scale_color_manual(values = c("grey", "black")) +
  scale_linetype_manual(values = c("solid", "dotted")) +
  scale_y_continuous(limits = c(-8, -1)) +
  theme_minimal() +
  theme(text = element_text(size = 14))
```

(A) English learning outcomes



```
knitr::opts_chunk$set(echo = TRUE)
```

```
#Dutch analysis
```

```
file_path <- "~/Desktop/Simultaneous_learning.csv"
data <- read.csv(file_path)
```

```
data$Time <- factor(data$Time)
data$Type <- factor(data$Target.Control)
data$Categorical_score <- factor(data$Categorical_score, ordered = TRUE, levels = 1:2)
data$Language <- factor(data$Language)
```

```
data <- subset(data, Language == "Dutch")
```

```
model_int <- glmer(
  Categorical_score ~ Time * Target.Control + English_proficiency + Dutch_proficiency +
    (1 | Participant) + (1 | ItemID),
  family = binomial, data = data
)
```

```
Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
```

Model failed to converge with max|grad| = 0.0693307 (tol = 0.002, component 1)

```
summary(model_int)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: Categorical_score ~ Time * Target.Control + English_proficiency +
Dutch_proficiency + (1 | Participant) + (1 | ItemID)
Data: data
```

AIC	BIC	logLik	deviance	df.resid
489.6	529.2	-236.8	473.6	1032

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.5315	-0.2551	-0.1212	-0.0469	8.3321

Random effects:

Groups	Name	Variance	Std.Dev.
ItemID	(Intercept)	1.877	1.370
Participant	(Intercept)	2.547	1.596

Number of obs: 1040, groups: ItemID, 40; Participant, 13

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-12.07221	4.34805	-2.776	0.0055 **
Time2	0.96548	0.59357	1.627	0.1038
Target.Control	0.60475	0.75367	0.802	0.4223
English_proficiency	0.03385	0.06517	0.519	0.6035
Dutch_proficiency	0.08389	0.05822	1.441	0.1496
Time2:Target.Control	1.10564	0.68697	1.609	0.1075

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	Time2	Trgt.C	Engls_	Dtch_p
Time2		-0.099			
Target.Cntrl	-0.119	0.529			
Englsh_prfc	-0.705	0.001	-0.003		
Dtch_prfcnc	-0.313	0.005	-0.003	-0.430	
Tm2:Trgt.Cn	0.045	-0.856	-0.621	0.009	0.023

optimizer (Nelder_Mead) convergence code: 0 (OK)

Model failed to converge with max|grad| = 0.0693307 (tol = 0.002, component 1)

```
r2_model_int <- performance::r2(model_int)
print(r2_model_int)
```

R2 for Mixed Models

Conditional R2: 0.665

Marginal R2: 0.213

```
emm <- emmeans(model_int, ~ Time * Target.Control, type = "response")
summary(emm)
```

Time	Target.Control	prob	SE	df	asyp.LCL	asyp.UCL
1	0	0.00517	0.00436	Inf	0.000988	0.0266
2	0	0.01348	0.01014	Inf	0.003054	0.0574
1	1	0.00943	0.00645	Inf	0.002455	0.0355
2	1	0.07025	0.03955	Inf	0.022536	0.1985

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

```
pairs(emm, by = "Target.Control") # change over Time within each group
```

Target.Control = 0:

contrast	odds.ratio	SE	df	null	z.ratio	p.value
Time1 / Time2	0.381	0.2260	Inf	1	-1.627	0.1038

Target.Control = 1:

contrast	odds.ratio	SE	df	null	z.ratio	p.value
Time1 / Time2	0.126	0.0447	Inf	1	-5.838	<.0001

Tests are performed on the log odds ratio scale

```
pairs(emm, by = "Time")
```

Time = 1:

contrast	odds.ratio	SE	df	null	z.ratio	p.value
Target.Control0 / Target.Control1	0.546	0.412	Inf	1	-0.802	0.4223

Time = 2:

contrast	odds.ratio	SE	df	null	z.ratio	p.value
Target.Control0 / Target.Control1	0.181	0.114	Inf	1	-2.716	0.0066

Tests are performed on the log odds ratio scale

```
library(ggplot2)
```

```
emm <- emmeans(model_int, ~ Time * Target.Control)
marginal_means_df <- as.data.frame(emm)
```

```
marginal_means_df$Target.Control <- factor(marginal_means_df$Target.Control, labels = c("Control", "Target"))
marginal_means_df$Time <- factor(marginal_means_df$Time,
                                levels = c(1, 2),
                                labels = c("Pretest", "Posttest"))
```

```
ggplot(marginal_means_df, aes(x = Time, y = emmean, color = Target.Control, group = Target.Control)) +
  geom_point(size = 3, position = position_dodge(width = 0.2)) +
  geom_line(aes(linetype = Target.Control), position = position_dodge(width = 0.2)) +
  geom_errorbar(aes(ymin = asymp.LCL, ymax = asymp.UCL),
               width = 0.1, position = position_dodge(width = 0.2)) +
```

```

labs(
  title = "(B) Dutch learning outcomes",
  x = "Time",
  y = "Estimated Marginal Means",
  color = "Condition",
  linetype = "Condition"
) +
scale_color_manual(values = c("grey", "black")) +
scale_linetype_manual(values = c("solid", "dotted")) +
scale_y_continuous(limits = c(-8, -1)) +
theme_minimal() +
theme(text = element_text(size = 14))

```

